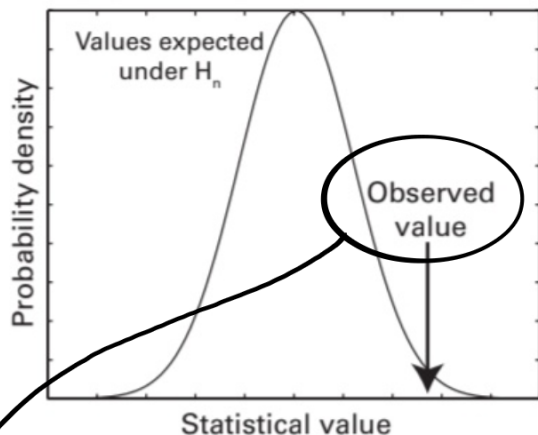


parametric vs Nonparametric:

A) Significance based on assumption



B) Significance based on data

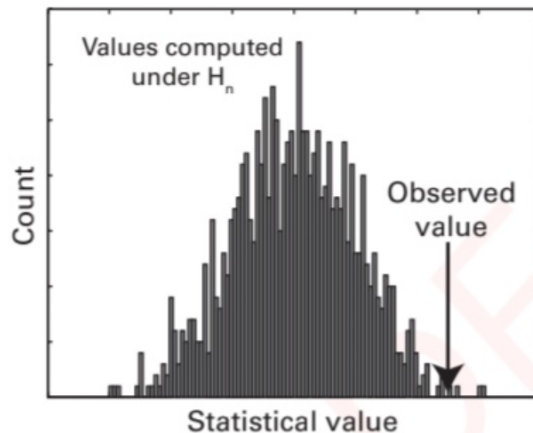


Figure 33.1

Illustration of distributions of test statistics under the null hypothesis (H_0) for permutation testing (panel A) and nonparametric permutation testing (panel B). The observed value is the value of the test statistic computed from the data without any adulterations or condition relabeling.

→ Test statistic computed from actual data we measured

↳ How do we know that it's big enough??

→ If that effect size is unlikely to occur given a distribution of Null hypothesis

↓
How do we get this distribution to compare against our observed values (2 ways)

Analytical solution

mathematic formula
(Assumptions about underlying population)

Based on Null Hypothesis
Empirical distribution
-ions → comes from permutation based approaches

↳ parametric

↳ computing parameters based on analytical formulas we have

↳ Non parametric

↳ because we are generating distributions based on actual data that we have

↳ simulating situations that could arise under the null hypothesis

Assumptions underlying parametric statistics

- Data sampled randomly from a population
- that population has known parameter distributions
- variance across repeated measurements is homogeneous
- Measurement error and sample variability are independent
- No bias
- parameters refers to key descriptive statistics like mean, variance, ...

Do Time Frequency data meet these Assumptions?

1) sampling neurons in electrophysiology

↳

↳ Not random

First assumption is sometimes met and sometimes not

- 2) Sometimes met and sometimes not, depending on analysis and some distributions can be transformed into known distributions (eg. dB power)
- 3) homogeneity $\rightarrow$ If we take repeated measurements of some system like brain, then variance across diff measurements should be same
 $\rightarrow$ Not satisfied
- 4) means, if we measure two diff individuals from population, they should be independent of each other
 $\rightarrow$ Sometimes yes and sometimes not

Assumptions Underlying Non parametric statistics

- $\rightarrow$ There exists a null hypothesis
- $\Rightarrow$ we have data from which to create a situation that would arise under the null hypothesis

Advantages of permutation based statistics over parametric based statistics:

- Don't need to worry about assumptions that are necessary for parametric statistics
- Ability to deal appropriately with Multiple Comparisons, in particular when our multiple comparisons has strong Auto correlation structure

Mechanism of permutation Testing:

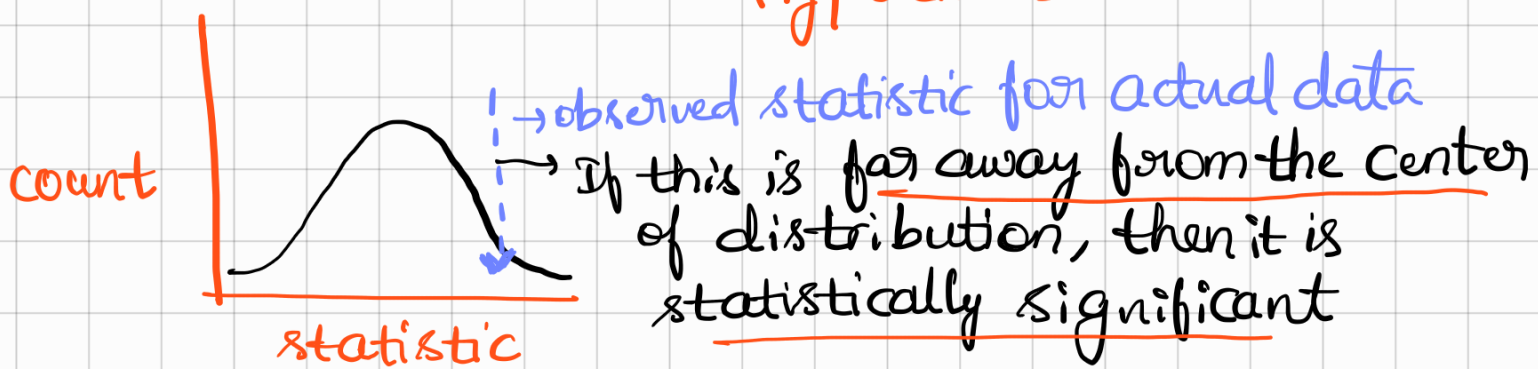
shuffle labels of conditions

compute test-statistic $\frac{\text{mean}_1 - \text{mean}_2}{\sigma_1 \sigma_2}$

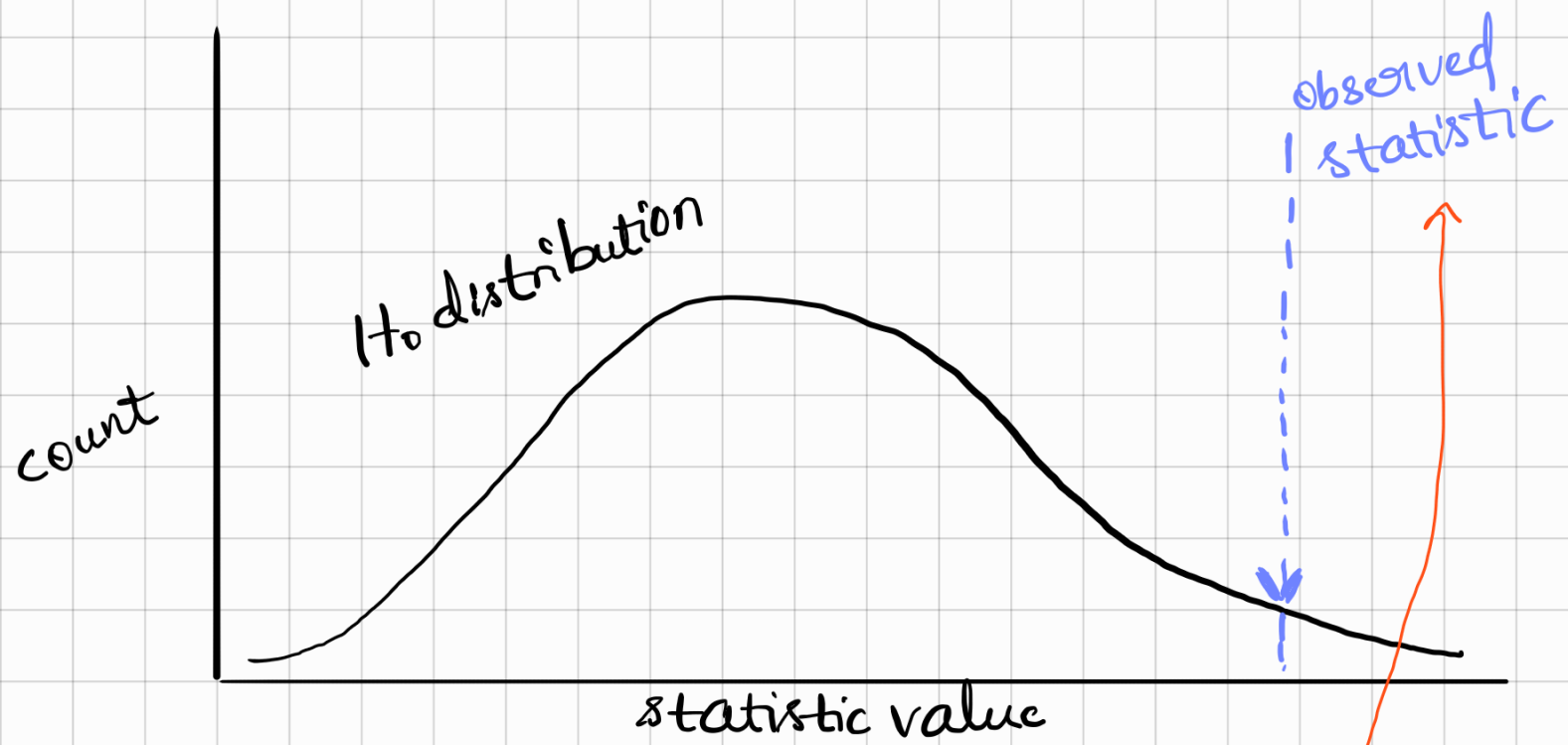
→ If Null hypothesis is true, then we expect the labels to be meaningless / random

reshuffle labels & compute statistic value
continue like 1000 times to build
distribution of statistic values

→ This is our Null Hypothesis distribution



Evaluating Significance (P-value)



2 ways of computing P-value:

1)
$$z = \frac{\text{obs} - E[H_0]}{\text{std}[H_0]}$$
 → mean of H_0 distribution

→ gives normalised distance in standard deviation units of observed statistic value away from the center of the Null Hypothesis distribution

→ larger the magnitude of the z -value, smaller the p-value

→ This formula is appropriate only for H_0 distributions that are approximately gaussian distributed

2) $p_c = \frac{\sum (H_0 > \text{obs})}{N_{H_0}} \rightarrow$ counting number of Null hypothesis statistics that are more extreme than observed test statistic value

$\rightarrow$ useful because it is robust to the shape of the distribution

$\rightarrow$ should be careful of which tail

How many Iterations is Enough:

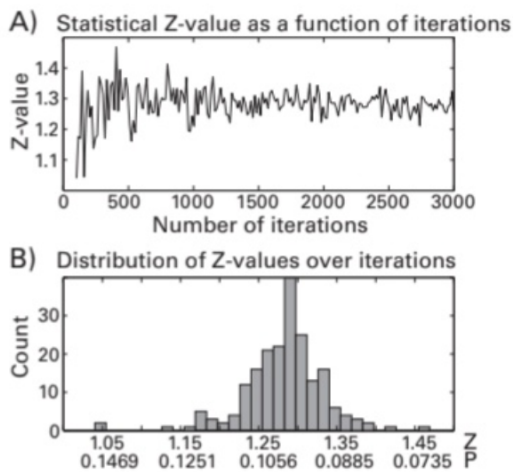


Figure 33.5

Panel A shows a test statistic value (y-axis, these are standard Z values from equation 33.1) as a function of the number of iterations used to generate the null hypothesis distribution (x-axis). Panel B shows a histogram of these test statistic values. The top row of x-axis labels shows Z-values, and the bottom row of x-axis labels shows the corresponding p-values.

at one iteration, we may get as significant & at other we may not, so how to get rid of this situation

$\hookrightarrow$ Meta permutation testing

Meta permutation test

- $\rightarrow$ we can see that Z-value never converges to particular value, but fluctuates around that particular value
- $\rightarrow$ Repeat permutation test multiple times & then average Z-value over all of those

individual permutation tests and that reflects the true underlying z -value

Coming to Time Frequency plot:

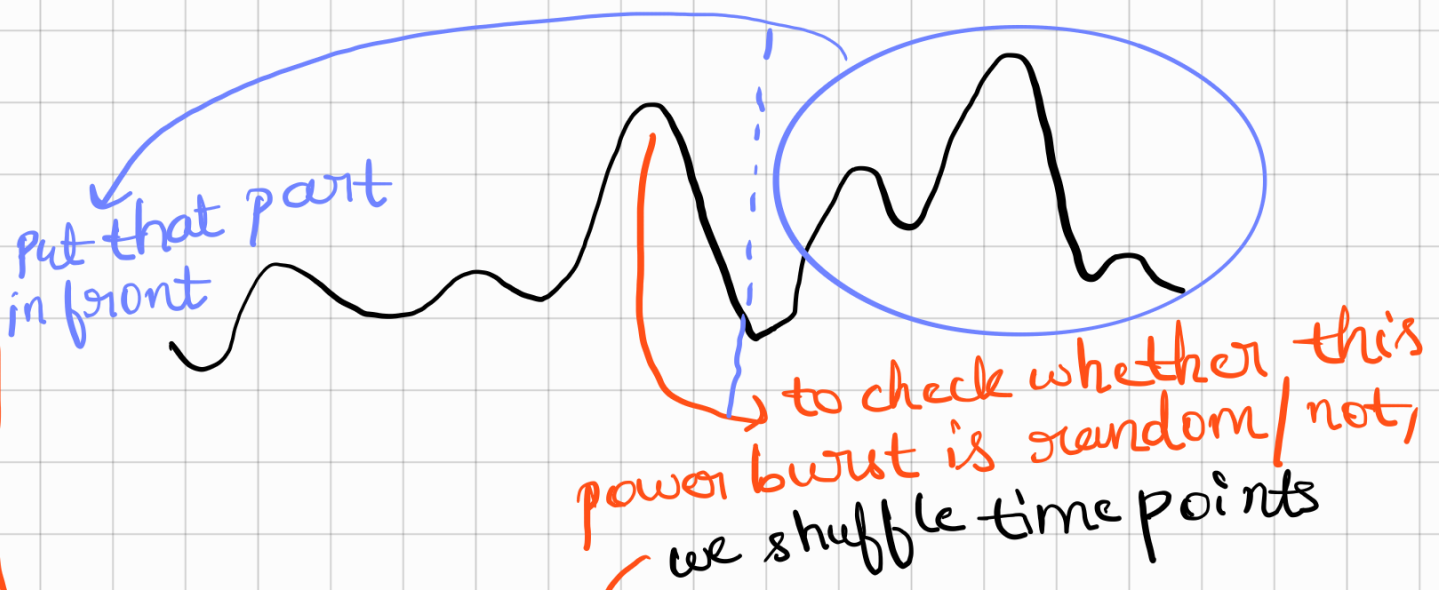
That above procedure is giving p -value for one pixel in Time Frequency plot and Each pixel in Time Frequency plot gets its own Null Hypothesis distribution

What to permute:

what is the feature of the data that we are going to permute

we should change the one key factor that our Null Hypothesis predicts doesn't matter

If we change too many factors at same time, we don't know what our results are attributable to



→ we have to manipulate exactly one feature of the data without changing anything else in data

→ Not appropriate way to shuffle data

instead randomly reassign time zero point, by doing this spectral structure of data won't be almost entirely unchanged

Multiple Comparisons and Limitations of Bonferroni:

lets say we have

1000 time points	} for one dataset
40 frequencies	
16 channels	
640,000 tests	

→ really large
lots of multiple comparisons

Bonferroni correction:

$$\text{Threshold} = \alpha / N$$

$$\alpha = \text{eg}; 0.05$$

$$N = \text{no. of tests}$$

problems with Bonferroni Correction :

- Too stringent
- Assumes independence
- Based on N , not on information in data
- we might have a bonferroni threshold of $0.05 / 640000$, which is tiny number, even real effects are unlikely to be significant using such a tiny threshold

→ Not the case for time frequency analysis, and also we apply smoothing on data for time frequency analysis, which violates independence

→ If we change any of no. of time points / frequencies in above example to lower, then p -value will be more liberal without changing any information content

→ For doing statistics on Time Frequency analysis results bonferroni correction is poor choice because of above problems

Appropriate Methods for Multiple Comparisons corrections of Time frequency data:

- 1) correct for clusters within the entire map
- 2) correct for extreme pixel values from the entire map

1) cluster correction:

Assumption: Activity is meaningful if it is in a "big enough" cluster (a cluster is a continuously significant region)

How big is "big enough"?

← No. of time points?
↘ No. of frequency bins?

this is arbitrary
(same arbitrary

→ solution: cluster size defined from data during permutation
means here we are going to let the data tell us what the appropriate cluster size should be

Mechanism of cluster correction:

one loop for creating permuted maps
other loop/iteration for getting clusters

lets say we got 1 map for iteration 1, this map was created assuming the null hypothesis is true, so we don't expect any pixels in map to be significant

There might be hundreds/thousands of data points inside the map, and if we are scanning uncorrected threshold at $p = 0.05$, it's very likely that we are going to get α -errors we assume all pixels that are less than our p value threshold

as alpha errors appearing by chance

we go through entire map and find largest cluster that occurred by chance, then we count no. of pixels in that cluster and similarly we do for all maps

and then we get distribution of cluster sizes and then we pick the 95th percentile in distribution

then we go back to original data, and look at all clusters, and for each cluster, if no. of pixels is greater than 95% above then keep that cluster, else remove that from map

cluster correction favours large clusters that means potentially meaningful real effects that are small in size on Time Frequency plot might be removed

Solution: Average T values of all pixels
(in clusters)
→ absolute value of T values

If we impose more smoothing on data, then clusters in real data will get larger, and also clusters in permutation testing will also get larger, so the threshold will also increase

2) Mechanism of Extreme-pixel Correction:

First loop is same

In second loop, we take most extreme negative and positive values from each map and we say in this entire map these are the two extreme values we can expect under the null hypothesis

→ Bimodal distribution

10% min and max values

we need to identify two separate thresholds

Two thresholds < $\begin{cases} 2.5\% \text{ on negative tail} \\ 97.5\% \text{ on positive tail} \end{cases}$

In original data map, any pixels with statistic value between thresholds get turned to zero and whatever left is significant

cluster correction vs pixel correction

retain entire cluster even though individual pixels are not significant
weakly significant

- Favours big / heavy clusters
- Sensible approach in Neuroscience
- Sensitivity can be frequency dependent

we don't pay any attention to whether the pixels are touching each other

- can detect small clusters / 1-pixel
- Interpretability depends on pixel size